

Heartgard®

(ivermectin)

Chewables for Cats

PRODUCT DESCRIPTION: Red to red-brown tinge, characteristic of fresh beef, no rancid, sour, putrid or other abnormal odors but rather a wholesome meaty odor.

PHARMACOLOGY: The tissue larval stage of *Dirofilaria immitis* is exquisitely sensitive to ivermectin. Since only a small amount of ivermectin is required to eliminate these larvae, the spectrum of activity of HEARTGARD for Cats against other parasites is limited as compared to the ivermectin formulations and doses used in other species. The action of the avermectins is unique and is not shared by other classes of antiparasitic agents. It involves the neurotransmitter gamma-aminobutyric acid, or GABA. Ivermectin stimulates the release of GABA from nerve endings and enhances binding of GABA to receptors at nerve junctions, thus interrupting nerve impulses and thereby paralyzing and killing the sensitive heartworm larvae. The recommended heartworm preventive dose of ivermectin has a wide safety margin in cats. The action of the principal peripheral neurotransmitter in mammals, acetylcholine, is unaffected by ivermectin. In mammals, ivermectin does not readily penetrate the central nervous system where GABA functions as a neurotransmitter.

The pharmacokinetics of ivermectin were evaluated in cats following administration at 24 mcg/kg b.w. using the oral and injectable (i.v.) routes in a two period crossover bioavailability study. After oral administration there was a rapid rise of plasma concentration, with a peak of 4.36 ng/ml reached in 5.5 hours, followed by a prolonged period in which ivermectin could be detected in plasma. Analysis of plasma levels following intravenous administration demonstrated that ivermectin has a mean biological half-life of 2.3 days in cats. The bioavailability of the oral formulation was approximately 25%.

INDICATION: For use in cats to prevent feline heartworm disease by eliminating the tissue stage of heartworm larvae (*Dirofilaria immitis*) for a month (30 days) after infection, and for the removal and control of adult and immature hookworms (*Ancylostoma tubaeforme* and *A. braziliense*).

DOSAGE: HEARTGARD for Cats should be administered orally at monthly intervals at the recommended minimum dose level of 24 mcg of ivermectin per kg (10.9 mcg/lb) of body weight. The recommended dosage schedule for prevention of feline heartworm disease and control of hookworm is as follows:

Cat Weight lb	HEARTGARD for Cats Chewables per Month	Ivermectin content
Up to 5 lb	1	55 mcg
5 to 15 lb	1	165 mcg

For cats over 15 lb, use the appropriate combination of chewables.

HEARTGARD for Cats is recommended for use in cats 6 weeks of age and older.

ADMINISTRATION: Remove only one chewable at a time from the foil-backed blister card. Return the card with the remaining chewables to its box to protect the product from light. HEARTGARD for Cats can be offered to the cat by hand or may be added to a small amount of cat food. If manual dosing is required, the chewable should be broken into pieces for administration. Food should be routinely available during the day of treatment administration. If cats are fasted, or if the chewable is not broken into pieces for manual dosing, then reduced absorption may result. A relationship between reduced absorption and reduced efficacy has not been established.

©HEARTGARD and the Cat & Hand logo are registered trademarks of Boehringer Ingelheim Animal Health USA Inc.

Care should be taken to see that the cat consumes the complete dose. Treated animals should be observed for a few minutes after administration to ensure that part of the dose is not lost or rejected. If it is suspected that any of the dose has been lost, redosing with a new chewable is recommended. Fragments of chewables have occasionally been observed in the feces of some cats; efficacy of the product was not adversely affected.

HEARTGARD for Cats should be given at monthly intervals when mosquitoes potentially carrying infective heartworm larvae are active. The initial dose must be given within a month (30 days) after the cat's first exposure to mosquitoes. The final dose must be given within a month (30 days) after the cat's last exposure to mosquitoes. For optimal performance, HEARTGARD for Cats must be given once a month on or about the same date. If treatment is delayed, whether by a few days or many, immediate treatment with HEARTGARD for Cats and resumption of the recommended dosing regimen will minimize the opportunity for development of adult heartworm.

It is recommended that cats should be tested for existing heartworm infection prior to starting treatment with HEARTGARD for Cats. Cats already infected with adult heartworms can be given HEARTGARD for Cats monthly to prevent further infections.

Monthly treatment with HEARTGARD® (ivermectin) for Cats also provides effective removal and control of adult and immature hookworms (*A. tubaeforme* and *A. braziliense*).

SAFETY: A wide margin of safety was demonstrated in clinical trials at the recommended dose level in cats and kittens 6 weeks of age and older and in laboratory studies in cats with circulating microfilariae of *D. immitis*. In the clinical studies involving more than 3000 doses of HEARTGARD for Cats, observations reported within 24 hours of treatment included vomition in ≤0.3% and diarrhea in ≤0.2% of the doses administered. There were no statistical differences between HEARTGARD for Cats and the product vehicle (control) for these observations.

A wide margin of safety has also been demonstrated at 3 times the recommended dose in pregnant or breeding queens and breeding toms. No adverse effects were observed in either male or female cats or their offspring in breeding studies.

A 30-fold safety margin over the minimum recommended dosage was established in a single dose tolerance study. In growing kittens dosed monthly for 8 consecutive doses, one cat given 5 times the market dose vomited and a second cat in this group experienced diarrhea within 24 hours of the initial treatment.

No chemical incompatibilities have been observed. In clinical trials, many commonly used flea control products, anthelmintics, vaccines, antibiotics and steroid preparations were administered with Heartgard for Cats without incident.

Keep this and all drugs out of the reach of children. In case of ingestion by humans, clients should be advised to contact a physician immediately. Used only when other treatment has failed.

HOW SUPPLIED: HEARTGARD for Cats is available in two dosage strengths (see DOSAGE section) for cats of different weights. Each strength comes in a convenient carton of 6 chewables, packed 10 cartons per tray.

For customer assistance, please contact Merial at 1-888-637-4251.

Manufactured by:
Boehringer Ingelheim Animal Health
Puerto Rico LLC,
Rd #2 KM 56.7, Barceloneta,
Puerto Rico 00617 USA

Repacker and registration holder:
Rhône Ma Malaysia Sdn Bhd (525887-W)
Lot 18A & 18B, Jalan 241, Seksyen 51A,
46100 Petaling Jaya,
Selangor, Malaysia

Product Owner:
Boehringer Ingelheim Vetmedica GmbH,
Binger Straße 173,
55216 Ingelheim am Rhein, Germany.

©2019 Boehringer Ingelheim Animal Health USA Inc.
All Rights Reserved.
Rev. 04-2019
1050-2169-06

Boehringer
Ingelheim

5LHEAPC3-006 04/XXX

LABEL CODE:		VERSION	COLORS
5LHEAPC3-006		Version: 4 MMM/YY:	BLACK
PRODUCT: HEARTGARD CHEWABLE FOR CATS			
PACK SIZE: 6 CHEWABLES			
COUNTRY: MY			
DIMENSIONS: cm = 10 x 20			

Marketing approval:		Regulatory approval:	GMP approval:
COMPLETE NAME:		COMPLETE NAME:	COMPLETE NAME:
DATE:		DATE:	DATE:
SIGNATURE:		SIGNATURE:	SIGNATURE: